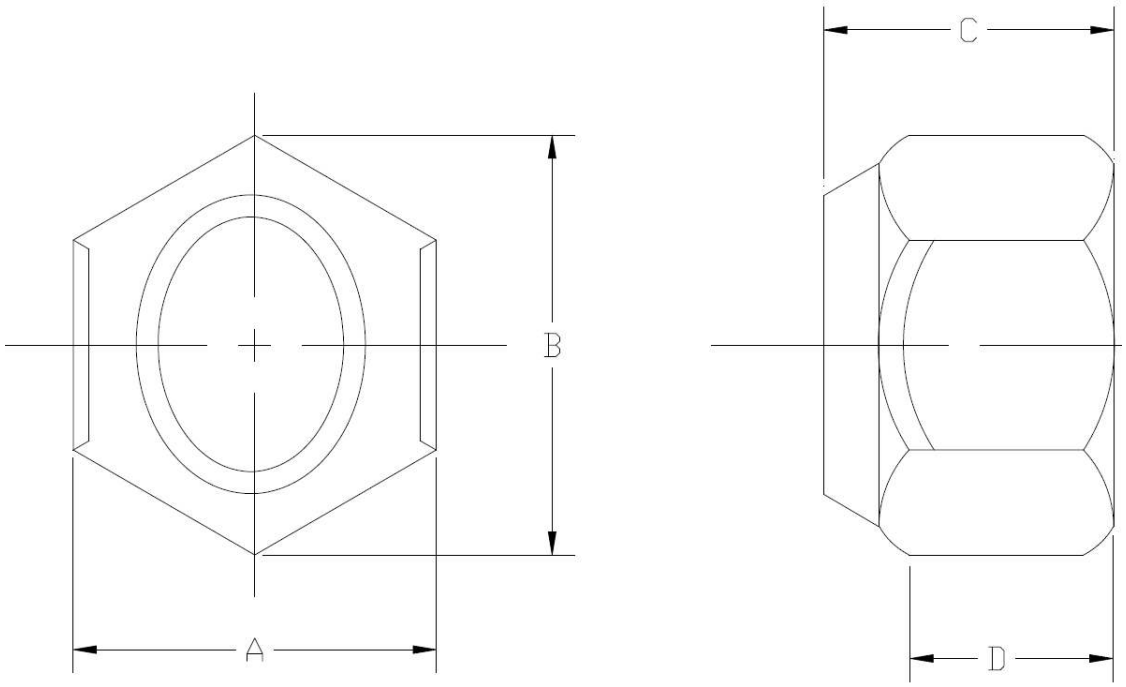
INCH-POUND
MIL-DTL-45913/2B
20 May 2021
SUPERSEDING
MIL-DTL-45913/2A
14 June 2011

# DETAIL SPECIFICATION SHEET

# NUT, SELF-LOCKING, HEXAGON
# ALL-METAL LOCKING FEATURE, 250 ºF, UNC/UNF-2B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-45913.




(All dimensions are in inches.)

FIGURE 1. Nut, self-locking, hexagon, all-metal locking feature, UNC/UNF-2B.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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REQUIREMENTS:

MATERIAL:

*Steel and alloy steel* shall be in accordance with SAE J995 grade 5 or 8, as specified in part-number.
*Corrosion resisting steel* (stainless steel) grades 303 and 316 or any alloy in accordance with ASTM F 594, Group 1 or Group 2, shall have a minimum ultimate tensile strength of 70 KSI and shall be non-magnetic.
*Copper alloy* (brass) shall be in accordance with SAE AMS4507, HR02 shall have a minimum ultimate tensile strength of 58 KSI and be of a half hard temper.
*Aluminum alloy* shall be 6061 in accordance with SAE AMS4117 and have a minimum ultimate tensile strength of 42 KSI or as specified in contract.

PLATING/FINISH:

Steel and alloy steel locknuts shall be:

Cadmium plated /1 per SAE-AMS-QQ-P-416 Type II, Class 2 or 3 or
Zinc-Nickel plated per MIL-PRF-32647, Type K16 or Type K20 or
Zinc plated per ASTM B633 /2 Type II, Fe/Zn 8 or
Black Oxide per MIL-DTL-13924, Class 1 or
Uncoated plain with a supplementary lubricant which is clean and dry to the touch.

Stainless steel locknuts shall be passivated SAE-AMS 2700 /3, Method 1, Type 6 or 7, Class 1, or black oxide finish per MIL-DTL-13924.

Brass locknuts shall be black oxide finished per MIL-F-495, nickel plated per SAE-AMS-QQ-N-290, Class 2, or uncoated plain with a brite finish.

Aluminum locknuts shall be anodized clear,, no color per MIL-PRF-8625 (see Table I), Class as specified in contract.

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred alternate.
/2 Type II passivation per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred alternates.
/3 Passivation per ASM2700 contains sodium dichromate which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred alternates.

SURFACE TEXTURE. Shall be Ra 125 microinches in accordance with ASME B46.1.

THREADS. Shall be coarse or fine threaded, class 2B, as specified in part-number, in accordance with ASME B1.1.

WORKMANSHIP. Parts shall be manufactured and processed with a level of care and workmanship befitting this type of product intended for general purpose military application. Parts shall be free from

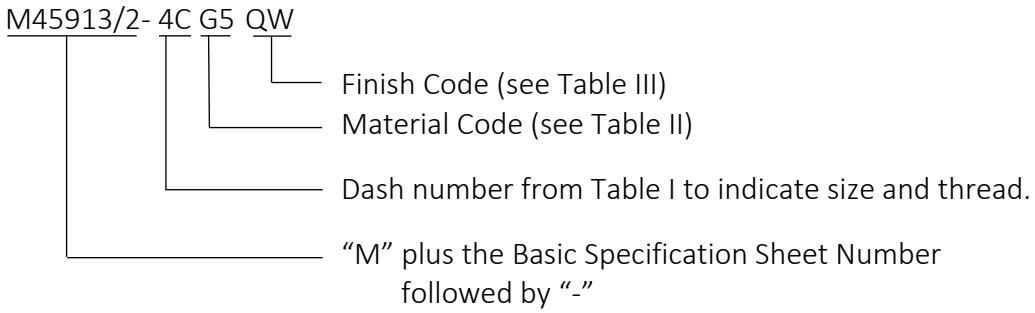


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dirt, grease, loose or deposited foreign materials, sharp edges, burrs, chips, physical malformations or anything which might prevent the part from reliably performing its intended function.

PART NUMBER. The part number shall consist of the basic number of this specification sheet followed by a dash, a dash number from I, a material code from Table II, and a finish/plate code (when required) from Table III.

EXAMPLE OF PART NUMBER:




M45913/2-4CG5QW = Nut Self-Locking Hexagon, all-metal locking feature, .250-20 thread, Grade 5 steel, Zinc-Nickel plated, Type K20.

TABLE I. Dash numbers and dimensions.

| Dash No. | Thread Size | A | | B | C | | D |
|---|---|---|---|---|---|---|---|
| | | Width Across Flats | | Corners | Thickness | | |
| | | MAX | MIN | REF | MAX | MIN | REF |
| 00F | .060-80 UNF-2B | .111 | .104 | .116 | .055 | .080 | .045 |
| 01C | .073-64 UNC-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 01F | .073-72 UNF-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 02C | .086-56 UNC-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 02F | .086-64 UNF-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 03C | .099-48 UNC-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 03F | .099-56 UNF-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 04C | .112-40 UNC-2B | .251 | .241 | .275 | .163 | .087 | .066 |
| 04F | .112-48 UNF-2B | .251 | .241 | .275 | .163 | .087 | .066 |
| 05C | .125-40 UNC-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 05F | .125-44 UNF-2B | .251 | .243 | .268 | .153 | .133 | .081 |
| 06C | .138-32 UNC-2B | .313 | .302 | .344 | .171 | .102 | .075 |
| 06F | .138-40 UNF-2B | .313 | .302 | .344 | .171 | .102 | .075 |
| 08C | .164-32 UNC-2B | .345 | .332 | .378 | .191 | .117 | .083 |
| 08F | .164-36 UNF-2B | .345 | .332 | .378 | .191 | .117 | .083 |
| 010C | .190-24 UNC-2B | .376 | .362 | .413 | .241 | .117 | .083 |
| 010F | .190-32 UNF-2B | .376 | .362 | .413 | .241 | .117 | .083 |
| 012C | .216-24 UNC-2B | .439 | .423 | .482 | .241 | .148 | .103 |
| 012F | .216-28 UNF-2B | .439 | .423 | .482 | .241 | .148 | .103 |
| 4C | .250-20 UNC-2B | .439 | .428 | .488 | .288 | .212 | .145 |
| 4F | .250-28 UNF-2B | .439 | .428 | .488 | .288 | .212 | .145 |
| 5C | .312-18 UNC-2B | .502 | .489 | .557 | .336 | .258 | .166 |
| 5F | .312-24 UNF-2B | .502 | .489 | .557 | .336 | .258 | .166 |



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TABLE I. Dash numbers and dimensions – Continued.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6C | .375-16 UNC-2B | .564 | .551 | .628 | .415 | .320 | .198 |
| 6F | .375-24 UNF-2B | .564 | .551 | .628 | .415 | .320 | .198 |
| 7C | .437-14 UNC-2B | .688 | .675 | .768 | .463 | .365 | .223 |
| 7F | .437-20 UNF-2B | .688 | .675 | .768 | .463 | .365 | .223 |
| 8C | .500-13 UNC-2B | .752 | .736 | .840 | .573 | .427 | .262 |
| 8F | .500-20 UNF-2B | .752 | .736 | .840 | .573 | .427 | .262 |
| 9C | .562-12 UNC-2B | .877 | .861 | .982 | .621 | .473 | .286 |
| 9F | .562-18 UNF-2B | .877 | .861 | .982 | .621 | .473 | .286 |
| 10C | .625-11 UNC-2B | .940 | .922 | 1.051 | .731 | .535 | .329 |
| 10F | .625-18 UNF-2B | .940 | .922 | 1.051 | .731 | .535 | .329 |
| 12C | .750-10 UNC-2B | 1.127 | 1.088 | 1.240 | .827 | .617 | .382 |
| 12F | .750-16 UNF-2B | 1.127 | 1.088 | 1.240 | .827 | .617 | .382 |
| 14C | .875-9UNC-2B | 1.314 | 1.269 | 1.447 | .922 | .724 | .450 |
| 14F | .875-14 UNF-2B | 1.314 | 1.269 | 1.447 | .922 | .724 | .450 |
| 16C | 1.000-8UNC-2B | 1.502 | 1.450 | 1.653 | 1.018 | .831 | .513 |
| 16F | 1.000-12 UNF-2B | 1.502 | 1.450 | 1.653 | 1.018 | .831 | .513 |
| 18C | 1.125-7UNC-2B | 1.689 | 1.631 | 1.859 | 1.176 | .939 | .576 |
| 18F | 1.125-12 UNF-2B | 1.689 | 1.631 | 1.859 | 1.176 | .939 | .576 |
| 20C | 1.250-7UNC-2B | 1.877 | 1.812 | 2.066 | 1.272 | 1.030 | .628 |
| 20F | 1.250-12 UNF-2B | 1.877 | 1.812 | 2.066 | 1.272 | 1.030 | .628 |
| 22C | 1.375-6UNC-2B | 2.064 | 1.994 | 2.273 | 1.399 | 1.138 | .681 |
| 22F | 1.375-12 UNF-2B | 2.064 | 1.994 | 2.273 | 1.399 | 1.138 | .681 |
| 24C | 1.500-6UNC-2B | 2.252 | 2.175 | 2.480 | 1.526 | 1.245 | .757 |
| 24F | 1.500-12 UNF-2B | 2.252 | 2.175 | 2.480 | 1.526 | 1.245 | .757 |

TABLE II. Material coding.

| MATERIAL CODE | |
|---|---|
| Steel Grade 5 | G5 |
| Steel Grade 8 | G8 |
| Stainless Steel 303 | S3 |
| Stainless Steel 316 | S6 |
| Brass | BB |
| Aluminum | AA |

TABLE III. Finish coding. 1/

| | |
|---|---|
| Cadmium | C |
| Zinc | Z |
| Black Oxide | B |
| Nickel | N |
| Zinc-Nickel (Type K16) | QX |
| Zinc-Nickel (Type K20) | QW |
| Anodized | no code (aluminum only) |
| Passivate | no code (stainless only) |
| Uncoated | no code |

(1/See requirements above for material and finish precautions.)



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NOTES:

1. Dimensions in inches, unless specified, tolerance: ± .016.

2. Interpret drawing in accordance with ASME Y14.5.

3. Break all sharp edged .003 to .005 and remove all burrs and slivers.

4. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

5. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.

6. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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